DA2403+DA2404 — Artificial Intelligence (+ lab)

Tier	A
Phase (order)	4
Do this first	DA2400, DA2303
Estimated effort	12 weeks
Unlocks	DA7400, reinforcement learning

Roadmap

1. Agents; search (BFS, A*), CSP, games.
2. Probabilistic models: HMMs, graphical models.
3. Planning and an intro to RL.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Artificial Intelligence Search Methods For Problem Solving (95 videos). NPTEL *AI: Search Methods for Problem Solving* (Deepak Khemani, IIT Madras).

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- [CS188 SP24] Lectures (27 videos). Berkeley CS188 *Intro to AI* (Spring 2024, official course channel): search, CSPs, games, MDPs, Bayes nets. Do the Pac-Man projects.
- MIT 6.034 Artificial Intelligence, Fall 2010 (30 videos). MIT 6.034 *Artificial Intelligence* (Patrick Winston, Fall 2010, official OCW).
- CS50's Introduction to Artificial Intelligence with Python 2023 (8 videos). Harvard CS50 AI (official): 7 lectures with Python projects, good for the lab (DA2404).

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–16	Philosophy and history of AI (watch at 1.5x).
2	17–22	State space search, DFS/BFS, iterative deepening. Implement them for the 8-puzzle.
3	23–33	Heuristic search, local search, TSP, simulated annealing, genetic algorithms, ant colony. Solve a TSP instance with hill climbing and a GA.

Week	Lectures	What to do
4-5	34–49	Optimal paths: branch & bound, A, *admissibility, monotone condition, memory-bounded variants. Implement A* on a grid; compare heuristics.
6	50–57	Game playing: minimax, alpha-beta, SSS*. Write a tic-tac-toe / Connect-4 AI with alpha-beta.
7-8	58–71	Planning: STRIPS, goal stack, plan-space, Graphplan, AO*. Write a blocks-world planner.
9	72–79	Rule-based systems, Rete algorithm.
10	80–87	Logic: inference, forward/backward chaining, FOL.
11	88–95	Constraint satisfaction, arc consistency, lookahead. Write a sudoku solver with AC-3 + backtracking.
12	Berkeley CS188 (alt)	Probabilistic half (Bayes nets, MDPs, RL) that this playlist does not cover. Do the Pac-Man projects.

▼ **All lectures in *Artificial Intelligence Search Methods For Problem Solving* (95 videos)**

1. Prologue
2. The Winograd Schema Challenge
3. Introduction (2013 version)
4. Can Machines Think?
5. The Turing Test
6. Language and Thought
7. The Willing Suspension of Disbelief
8. Machines with Wheels and Gears
9. The Notion of Mind in Philosophy
10. Reasoning = Computation
11. Concepts and Categories
12. How did AI get its name?
13. The Chess Saga
14. A Brief History of AI
15. The Worlds in our Minds
16. Epiphemona in Computers
17. State Space Search
18. Domain Independent Algorithms
19. Deterministic Search
20. DFS and BFS
21. Comparing DFS and BFS
22. Depth First Iterative Deepening
23. Heuristic Search
24. Heuristic Functions and the Search Landscape

25. Solution Space Search
26. The Traveling Salesman Problem
27. Escaping Local Optima
28. Stochastic Local Search
29. Genetic Algorithms: Survival of the Fittest
30. Genetic Algorithms and SAT
31. Genetic Algorithms for the TSP
32. Emergent Systems
33. Ant Colony Optimization
34. Finding Optimal Paths
35. Branch & Bound
36. Algorithm A*
37. A*: An illustrated example
38. Is A* Admissible?
39. Admissibility of A*
40. Higher, Faster ...
41. B&B - A - *wA* - Best First
42. A*: Leaner Admissible Variations
43. The Monotone Condition
44. DNA Sequence Alignment
45. Divide & Conquer Frontier Search.
46. Smart Memory Graph Search
47. Variations on A*: The story so far
48. Breadth First Heuristic Search
49. Beam Stack Search
50. Game Theory
51. Popular Recreational Games
52. Board Games and Game Trees
53. The Evaluation Function in Board Games
54. Algorithm Minimax and Alpha-Beta Pruning
55. A Cluster of Strategies
56. SSS*: A Best First Algorithm
57. SSS*: A Detailed Example
58. Automated Domain Independent Planning
59. The Blocks World Domain
60. State Space Planning: Forward and Backward
61. Goal Stack Planning (GSP)
62. GSP: A Detailed Example
63. Plan Space Planning (PSP)
64. PSP: A Tiny Example
65. Multi-Armed Robots

66. Means-Ends Analysis
67. The Planning Graph
68. Algorithm Graphplan
69. Problem Decomposition.
70. Algorithm AO*
71. AO*: An Illustration
72. Rule Based Expert Systems
73. The Inference Engine
74. The OPS5 Language
75. Conflict Resolution
76. Business Rule Management Systems
77. The Rete Net
78. Rete Algorithm: Optimizing the Match
79. Rete Algorithm: Conflict Resolution
80. Reasoning in Logic
81. Rules of Inference
82. Forward Reasoning
83. First Order Logic
84. Implicit Quantifier Notation
85. Backward Reasoning
86. Depth First Search on Goal Trees
87. Incompleteness...
88. Constraint Satisfaction Problems
89. Binary Constraint Networks
90. Interpreting Line Drawings
91. Model Based Diagnosis
92. Solving CSPs
93. Arc Consistency
94. Propagation = Reasoning
95. Lookahead Search

Books, notes and websites

- Book: Russell & Norvig, *AIMA* (the course text); Sutton & Barto; Bishop
- ✅ NPTEL *AI: Search Methods* (Khemani, IIT Madras)
- Berkeley CS188

Practice / proof of understanding

Solve Pacman-style search, CSP and an HMM filter in Python.

What your IITM course sheet says

From the PDF: DA2403

Official description: The students should be able to 1. Formulate the given AI task at hand in the rational agents and environments framework. 2. Construct basic algorithms for the most common type of AI tasks. 3. Implement these algorithms on the computer for a variety of settings.

Official syllabus (trimmed):

> 1. Introduction (2 weeks). Strong and weak AI, rational agents and environments, descriptive, predictive and prescriptive/control tasks, multi-agent cooperative and adversarial tasks.
> 2. Description, Prediction, Modelling and Decisions (3 weeks) Clustering, prediction, classification vs regression, sequential prediction, hidden Markov models, probabilistic graphical models: inference and learning.
> 3. Search (4 weeks) Deterministic control, single and multiple agents, BFS, DFS, uniform cost, A*, heuristics, constraint satisfaction problems. Games, adversarial search, minimax value, a - ß pruning.
> 4. Logic and Planning (3 weeks) Logical agents, first-order logic, planning, knowledge representation.
> 5. Stochastic Control (2 weeks) Markov Decision Processes, formulation, value Iteration, Q-learning, partially obsevable MDPs, Reinforcement learning.
> 6. Advanced Topics (1 week) Generative AI, robotics, real world impact. Fairness, accountability and transparency in AI.

Official textbooks: 1. Artificial Intelligence : A Modern Approach (Paperpack). Stuart Russell and Peter Norvig. Pearson; 3 edition. 2010 ISBN-13: 978-0132071482 2. Reinforcement Learning: An Introduction. Richard S. Sutton Andrew G. Barto . MIT Press, 2017. ISBN-13: 9780262332767 3. Pattern Recognition and Machine Learning. Christopher Bishop. Springer. 2006. ISBN-13 978-0-387-31073-2.

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From the PDF: DA2404

Official description: The students should be able to 1. Construct basic AI algorithms for the most common type of AI tasks. 2. Implement these algorithms on the computer for a variety of settings.

Official syllabus (trimmed):

> 1. Description, Prediction, Modelling and Decisions (3 weeks) Clustering, prediction, classification vs regres- sion, sequential prediction, hidden Markov models, probabilistic graphical models: inference and learning.
> 2. Search (4 weeks) Deterministic control, single and multiple agents, BFS, DFS, uniform cost, A*, heuristics, constraint satisfaction problems. Games, adversarial search, minimax value, a - ß pruning.
> 3. Stochastic Control (2 weeks) Markov Decision Processes, formulation, value Iteration, Q-learning, partially obsevable MDPs.
> 4. Mini-projects (5 weeks) Develop AI based solutions to complex problems on various environments. Build game solvers.

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